MH4110 PDE

Tutorial 09

Question 1

Solve the diffusion problem $u_t = ku_{xx}$ in $-l < x < 0$, with the boundary conditions $u_x(-l, t) = u_x(0, t) = 0$ and initial condition $u(x, 0) = \phi(x)$.

Question 1

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We use the method of separation of variables to solve this problem. First, we look for a solution of the form: $u(x, t) = X(x)T(t)$. In this case, we get

$$-\frac{X''}{X} = -\frac{T'}{kT} := \lambda.$$

λ must be a constant. Thus, we have the eigen-problem as

$$X'' = -\lambda X, \quad X'(-l) = X'(0) = 0.$$

(1) If $\lambda < 0$, then $X(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$ with c_1 and c_2 as arbitrary constants. Thus, $X'(x) = -c_1\sqrt{-\lambda}e^{-\sqrt{-\lambda}x} + c_2\sqrt{-\lambda}e^{\sqrt{-\lambda}x}$. The boundary conditions $X'(-l) = X'(0) = 0$ imply that $c_1 = c_2 = 0$. The solution is trivial.

(2) If $\lambda = 0$, $X''(x) = 0$ and then $X(x) = ax + b$. The boundary conditions $X'(-l) = X'(0) = 0$ imply that $a = 0$ and $X(x) = b$ is a constant.

(3) If $\lambda > 0$, then $X(x) = C\cos(\sqrt{\lambda}x) + D\sin(\sqrt{\lambda}x)$ with C and D as arbitrary constants. Thus, $X'(x) = -C\sqrt{\lambda}\sin(\sqrt{\lambda}x) + D\sqrt{\lambda}\cos(\sqrt{\lambda}x)$. The boundary condition $X'(0) = 0$ gives $D = 0$. Then, the boundary condition $X'(-l) = 0$ suggests that $\sqrt{\lambda}l = n\pi$.

$$\lambda = \frac{n^2\pi^2}{l}$$

Question 1

(a, b)

Solve the diffusion problem $u_t = ku_{xx}$ in $-l < x < 0$, with the boundary conditions $u_x(-l, t) = u_x(0, t) = 0$ and initial condition $u(x, 0) = \phi(x)$.

$u_x(a, t) = u_x(b, t)$

So, the eigenproblem admits the eigen-pairs

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \cos\frac{n\pi x}{l}, \quad n = 0, 1, 2, \cdots.$$

$= 0$

Solve $T' + \lambda k T = 0$, and obtain

$$T_n(t) = A_n \exp\left(-\left(\frac{n\pi}{l}\right)^2 kt\right).$$

The solution is

$$u(x, t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \exp\left(-\left(\frac{n\pi}{l}\right)^2 kt\right) \cos\frac{n\pi x}{l}.$$

In view of $u(x, 0) = \phi(x)$, with A_n to be determined we deduce that

$$\phi(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l}.$$

Question 2

Consider the dynamics of a vibrating string with infinite length in the presence of an external force

$$\begin{cases} u_{tt} - c^2 u_{xx} = f(x,t), & \text{for } -\infty < x < \infty, t > 0, \\ u(x,0) = 0, \quad u_t(x,0) = 0 \end{cases}$$

where

$$f(x,t) = \begin{cases} a, & \text{for } -a < x < a, \\ 0, & \text{otherwise.} \end{cases}$$

Find the position of the point $x = 0$ on the string, $u(0,t)$, at any time t.

Question 2

Consider the dynamics of a vibrating string with infinite length in the presence of an external force

$$\begin{cases} u_{tt} - c^2 u_{xx} = f(x,t), & \text{for } -\infty < x < \infty, t > 0, \\ u(x,0) = 0, \quad u_t(x,0) = 0 \end{cases}$$

where

$$f(x,t) = \begin{cases} a, & \text{for } -a < x < a, \\ 0, & \text{otherwise.} \end{cases}$$

Find the position of the point $x = 0$ on the string, $u(0,t)$, at any time t.

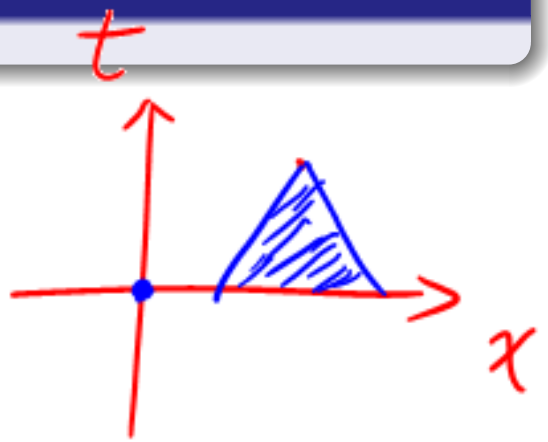
For the inhomogeneous wave equation, we have the solution formula

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy$$
$$+ \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds.$$

Question 2

Substituting $\phi(x) = 0$ and $\psi(x) = 0$ into the above formula gives

$$u(x,t) = \frac{1}{2}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds.$$



We are interested in $u(0,t)$; that is

$$u(0,t) = \frac{1}{2c}\int_0^t \int_{-c(t-s)}^{c(t-s)} f(y,s)dyds.$$

Given that

$$f(x,t) = \begin{cases} a, & \text{for } -a < x < a, \\ 0, & \text{otherwise,} \end{cases}$$

$(-c(t-s), c(t-s))$

Case 1

Consider $t < \frac{a}{c}$, we have $c(t-s) < a$ and $f(x,\ t) = a$; thus,

$$u(0,t) = \frac{1}{2c}\int_0^t \int_{-c(t-s)}^{c(t-s)} f(y,s)dyds = \frac{1}{2c}\int_0^t \int_{-c(t-s)}^{c(t-s)} adyds = \frac{1}{2c}\int_0^t 2ac(t-s)ds = \frac{a}{2}t^2.$$

Consider $t \geq \frac{a}{c}$, we have $c(t-s) \geq a$ when $s \leq t - \frac{a}{c}$; thus,

$\rightarrow c(t-s) \leq a$

$$u(0,t) = \frac{1}{2c}\int_0^{t-\frac{a}{c}} \int_{-c(t-s)}^{\overset{\geq a}{c(t-s)}} f(y,s)dyds + \frac{1}{2c}\int_{t-\frac{a}{c}}^{t} \int_{-c(t-s)}^{c(t-s)} f(y,s)dyds$$

$c(t-s) > a$

$$= \frac{1}{2c}\int_0^{t-\frac{a}{c}} \int_{-a}^{a} f(y,s)dyds + \frac{1}{2c}\int_{t-\frac{a}{c}}^{t} \int_{-c(t-s)}^{c(t-s)} f(y,s)dyds$$

$\Rightarrow t > \frac{a}{c} + s$

Question 2

Consider $t < \frac{a}{c}$, we have $c(t-s) < a$ and $f(x,\, t) = a$; thus,

$$u(0,t) = \frac{1}{2c}\int_0^t \int_{-c(t-s)}^{c(t-s)} f(y,s)dyds = \frac{1}{2c}\int_0^t \int_{-c(t-s)}^{c(t-s)} adyds = \frac{1}{2c}\int_0^t 2ac(t-s)ds = \frac{a}{2}t^2.$$

Consider $t \geq \frac{a}{c}$, we have $c(t-s) \geq a$ when $s \leq t - \frac{a}{c}$; thus,

$$\begin{aligned}
u(0,t) &= \frac{1}{2c}\int_0^{t-\frac{a}{c}} \int_{-c(t-s)}^{c(t-s)} f(y,s)dyds + \frac{1}{2c}\int_{t-\frac{a}{c}}^t \int_{-c(t-s)}^{c(t-s)} f(y,s)dyds \\
&= \frac{1}{2c}\int_0^{t-\frac{a}{c}} \int_{-a}^{a} f(y,s)dyds + \frac{1}{2c}\int_{t-\frac{a}{c}}^t \int_{-c(t-s)}^{c(t-s)} f(y,s)dyds \\
&= \underline{\frac{1}{2c}\int_0^{t-\frac{a}{c}} \int_{-a}^{a} adyds + \frac{1}{2c}\int_{t-\frac{a}{c}}^t \int_{-c(t-s)}^{c(t-s)} adyds} \\
&= \frac{a^2}{c}\left(t - \frac{a}{c}\right) + \frac{a^3}{2c^2}
\end{aligned}$$

Thus,

$$u(0,t) = \begin{cases} \frac{a}{2}t^2, & t < \frac{a}{c}, \\ \frac{a^2}{c}t - \frac{a^3}{2c^2}, & t \geq \frac{a}{c}. \end{cases}$$

Question 3

Consider waves that satisfy the problem

$$\begin{cases} u_{tt} = c^2 u_{xx}, & 0 < x < l,\ t > 0, \\ u_x(0,t) = u(l,t) = 0, & t > 0, \\ u(x,0) = \phi(x),\ u_t(x,0) = \psi(x), & 0 < x < l. \end{cases}$$

Write down the series expansion of the solution.

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(a, b)

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Write down the series expansion of the solution.

Look for the solution: $u(x,t) = X(x)T(t)$. Then we have

$$XT'' = c^2 X'' T.$$

Dividing both sides by $-c^2 XT$, gives

$$-\frac{T''}{c^2 T} = -\frac{X''}{X} := \lambda.$$

This yields the eigen-problem:

$$X'' + \lambda X = 0, \quad X'(0) = X(l) = 0,$$

and the ODE for T :

$$T'' + c^2 \lambda T = 0.$$

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If $\lambda = 0$, $X''(x) = 0$ and then $X(x) = ax + b$. The boundary conditions $X'(0) = X(l) = 0$ imply that $a = 0$ and $b = 0$. The solution is trivial.
If $\lambda > 0$, then $X(x) = C\cos(\sqrt{\lambda}x) + D\sin(\sqrt{\lambda}x)$ with C and D as arbitrary constants. Thus, $X'(x) = -C\sqrt{\lambda}\sin(\sqrt{\lambda}x) + D\sqrt{\lambda}\cos(\sqrt{\lambda}x)$. The boundary condition $X'(0) = 0$ gives $D = 0$. Then, the boundary condition $X(l) = 0$ suggests that $C\cos(\sqrt{\lambda}l) = 0$ and $\sqrt{\lambda}l = (n + \frac{1}{2})\pi$.

The eigenvalues and eigenfunctions are

$$\lambda_n = \frac{\left(n + \frac{1}{2}\right)^2 \pi^2}{l^2}, \quad X_n(x) = \cos\left(\frac{(n + \frac{1}{2})\pi x}{l}\right), \quad n = 0, 1, 2, \cdots.$$

We turn to the ODE for T :

$$T_n(t) = (A_n \cos\beta_n t + B_n \sin\beta_n t), \quad \beta_n = \frac{c(n + \frac{1}{2})\pi}{l}.$$

Question 3

The solution formula is

$$u(x,t) = \sum_{n=0}^{\infty} \left(A_n \cos \beta_n t + B_n \sin \beta_n t\right) \cos\left(\frac{(n+\frac{1}{2})\pi x}{l}\right),$$

where A_n and B_n are determined by the initial conditions as

$$\phi(x) = u(x,0) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{(n+\frac{1}{2})\pi x}{l}\right),$$

and

$$\psi(x) = u_t(x,0) = \sum_{n=0}^{\infty} B_n \frac{c(n+\frac{1}{2})\pi}{l} \cos\left(\frac{(n+\frac{1}{2})\pi x}{l}\right).$$

Question 4

Consider waves that satisfy the problem

$$\begin{cases} u_{tt} = c^2 u_{xx}, & -l < x < l,\ t > 0, \\ u(-l,t) = u(l,t) = 0, & t > 0, \\ u(x,0) = \phi(x),\ u_t(x,0) = \psi(x), & -l < x < l. \end{cases}$$

Write down the series expansion of the solution.

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Write down the series expansion of the solution.

Look for the solution: $u(x,t) = X(x)T(t)$. Then we have

$$XT'' = c^2 X'' T.$$

Dividing both sides by $-c^2 XT$, gives

$$-\frac{T''}{c^2 T} = -\frac{X''}{X} := \lambda.$$

This yields the eigen-problem:

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If $\lambda > 0$, then $X(x) = C\cos(\sqrt{\lambda}x) + D\sin(\sqrt{\lambda}x)$ with C and D as arbitrary constants. The boundary conditions $X(-l) = X(l) = 0$ give (1) $D = 0$ and $\sqrt{\lambda}l = (n + \frac{1}{2})\pi$ with nonnegative integer $n \geq 0$ or (2) $C = 0$ and $\sqrt{\lambda}l = n\pi$ with postive integer $n > 0$.

$$\begin{cases} C\cos\sqrt{\lambda}\,l - D\sin\sqrt{\lambda}\,l = 0 \\ C\cos\sqrt{\lambda}\,l + D\sin\sqrt{\lambda}\,l = 0 \end{cases} \Leftarrow \quad C\cos\sqrt{\lambda}\,l = 0 \text{ and } D\sin\sqrt{\lambda}\,l = 0$$

Question 4

(1) We consider the first case as $D = 0$ and $\sqrt{\lambda} l = (n + \frac{1}{2})\pi$ with nonnegative integer $n \geq 0$. The eigenvalues and eigenfunctions are

$$\lambda_n = \frac{\left(n + \frac{1}{2}\right)^2 \pi^2}{l^2}, \quad X_n(x) = \cos\left(\frac{(n + \frac{1}{2})\pi x}{l}\right), \quad n = 0, 1, 2, \cdots.$$

We turn to the ODE for T.

$$T_n(t) = \left(A_n \cos \beta_n t + B_n \sin \beta_n t\right), \quad \beta_n = \frac{c(n + \frac{1}{2})\pi}{l}.$$

The solution formula is

$$u(x, t) = \sum_{n=0}^{\infty} \left(A_n \cos \beta_n t + B_n \sin \beta_n t\right) \cos\left(\frac{(n + \frac{1}{2})\pi x}{l}\right),$$

where A_n and B_n are determined by the initial conditions as

$$\phi(x) = u(x, 0) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{(n + \frac{1}{2})\pi x}{l}\right),$$

and

$$\psi(x) = u_t(x, 0) = \sum_{n=0}^{\infty} B_n \frac{c(n + \frac{1}{2})\pi}{l} \cos\left(\frac{(n + \frac{1}{2})\pi x}{l}\right).$$

Question 4

(2) Now we consider the second case as $C = 0$ and $\sqrt{\lambda} l = n\pi$ with positive integer $n > 0$. The eigenvalues and eigenfunctions are

$$\lambda_n = \frac{n^2\pi^2}{l^2}, \quad X_n(x) = \sin\left(\frac{n\pi x}{l}\right), \quad n = 1, 2, \cdots.$$

We turn to the ODE for T :

$$T_n(t) = \left(A_n \cos\beta_n t + B_n \sin\beta_n t\right), \quad \beta_n = \frac{nc\pi}{l}.$$

The solution formula is

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos\beta_n t + B_n \sin\beta_n t\right) \sin\left(\frac{n\pi x}{l}\right),$$

where A_n and B_n are determined by the initial conditions as

$$\phi(x) = u(x,0) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{l}\right),$$

and

$$\psi(x) = u_t(x,0) = \sum_{n=1}^{\infty} B_n \frac{nc\pi}{l} \sin\left(\frac{n\pi x}{l}\right).$$

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Tutorial 10

§5.4 Completeness

We state the basic theorems about the convergence of Fourier series in §5.4.

Three notions of convergence

(i) We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ **pointwisely**, if for each $x \in (a, b)$,

$$\lim_{N\to\infty} S_N(x) = f(x).$$

(ii) We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ **uniformly**, if

$$\lim_{N\to\infty} \max_{x\in[a,b]} |f(x) - S_N(x)| = 0.$$

(iii) We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ in the L^2**-sense** (or the **mean-square sense**), if

$$\lim_{N\to\infty} \int_a^b |f(x) - S_N(x)|^2 dx = 0.$$

§5.4 Completeness

Three notions of convergence (relations)

(i) Uniform convergence is **stronger** than the other two notions of convergence;

(ii) Pointwise convergence and the L^2-convergence do not imply each other.

Theorem 2. Uniform Convergence

The Fourier (sine, cosine or full) series of $f(x)$, converges uniformly to $f(x)$ provided that

(i) for Fourier sine series: $f(x)$ and $f'(x)$ are continuous on $[0, l]$ and $f(0) = f(l) = 0$;

(ii) for Fourier cosine series: $f(x)$ and $f'(x)$ are continuous on $[0, l]$;

(iii) for full Fourier series: $f(x)$ and $f'(x)$ are continuous on $[-l, l]$ and $f(-l) = f(l)$.

§5.4 Completeness

Theorem 3. L^2-Convergence

The Fourier series converges to $f(x)$ in the mean-square sense in (a, b) provided only that $f(x)$ is any function for which

$$\int_a^b |f(x)|^2 dx$$

is finite, i.e., $f \in L^2(a, b)$.

Theorem 3. L^2-Convergence

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is finite, i.e., $f \in L^2(a, b)$.

Remark: The condition is weak, which means the Fourier series of a large class of functions converge in L^2-sense. Notice that all continuous functions are in L^2.

§5.4 Completeness

Theorem 4. Pointwise Convergence of Classical Fourier Series

(i) The classical Fourier series (full or sine or cosine) converges to $f(x)$ pointwisely on (a, b) provided that $f(x)$ is a continuous function on $[a, b]$, and $f'(x)$ is piecewise continuous on $[a, b]$.

(ii) More generally, if f itself is only piecewise continuous on $[a, b]$ and $f'(x)$ is also piecewise continuous on $[a, b]$, then the Fourier series converges at every point $x \in (-\infty, \infty)$. Moreover, we have the convergence:

$$\sum_n A_n X_n(x) = \frac{1}{2}\big[f(x+) + f(x-)\big], \quad -\infty < x < \infty, \tag{1}$$

where $f(a\pm) = \lim_{x \to a^\pm} f(x)$. Here, $f(x)$ is the extension (periodic, odd periodic or even periodic) of f in the original interval, e.g., $(0, l)$ or $(-l, l)$.

Question 1

Use the method of separation of variables to solve the boundary value problem for the diffusion equation

$$\begin{cases} u_t = ku_{xx}, \quad \text{for } 0 < x < l, \quad t > 0, \\ u_x(0,t) = u_x(l,t) = 0, \\ u(x,0) = x + 2. \end{cases}$$

Question 1

The diffusion equation with the Neumann BCs has the solution

$$u(x,t) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n e^{-(n\pi/l)^2 kt} \cos\frac{n\pi x}{l}.$$

Put $t = 0$, we have

$$x + 2 = u(x,0) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l}.$$

Assume that

$$x + 2 = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l},$$

we need to evaluate

$$A_m = \frac{2}{l}\int_0^l (x+2)\cos\frac{m\pi x}{l}dx, \quad m = 0, 1, 2, \cdots, \tag{2}$$

by integration by parts. More precisely,

$$A_0 = \frac{2}{l}\int_0^l (x+2)dx = 4 + l.$$

Question 1

$$A_n = \frac{2}{l}\int_0^l (x+2)\cos\frac{n\pi x}{l}\,dx = \frac{2}{n\pi}\int_0^l (x+2)\,d\sin\frac{n\pi x}{l}$$
$$= \frac{2}{n\pi}(x+2)\sin\frac{n\pi x}{l}\Big|_0^l - \frac{2}{n\pi}\int_0^l \sin\frac{n\pi x}{l}\,dx$$
$$= \frac{2l}{(n\pi)^2}\cos\frac{n\pi x}{l}\Big|_0^l = \frac{2l}{(n\pi)^2}\left[\cos(n\pi)-1\right] = \frac{2l}{(n\pi)^2}\left[(-1)^n-1\right].$$

We have the Fourier cosine expansion:

$A_0/2$

$$x+2 = 2+\frac{l}{2}+\frac{2l}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n-1}{n^2}\cos\frac{n\pi x}{l}, \quad x\in(0,l).$$

The solution is therefore

$$u(x,t) = 2+\frac{l}{2}+\sum_{n=1}^{\infty}\frac{2l}{(n\pi)^2}\left[(-1)^n-1\right]e^{-(n\pi/l)^2kt}\cos\frac{n\pi x}{l}.$$

$$= \begin{cases} 2+l/2 & n=2m \quad m\geq 1 \\ 2+\frac{l}{2}-\sum_{m=1}^{\infty}\frac{4l}{(2m-1)^2\pi^2}e^{-\left(\frac{2m-1}{l}\pi\right)^2kt}\cos\frac{(2k-1)\pi x}{l} \end{cases}$$

Question 2

$T(t)$ is a function of the single variable t and it satisfies

$$T'' + 2T' + \lambda T = 0.$$

Find the general solution of $T(t)$.

Question 2

$T(t)$ is a function of the single variable t and it satisfies

$$T'' + 2T' + \lambda T = 0.$$

Find the general solution of $T(t)$.

The characteristic function is

$$x^2 + 2x + \lambda = 0.$$

To find the general solution of $T(t)$, we need to check the sign of $\Delta = 4 - 4\lambda$.

When $\Delta = 4 - 4\lambda > 0$, i.e. $\lambda < 1$, the general solution is

$$T(t) = c_1 e^{\left(-1-\sqrt{1-\lambda}\right)t} + c_2 e^{\left(-1+\sqrt{1-\lambda}\right)t}.$$

When $\Delta = 4 - 4\lambda = 0$, i.e. $\lambda = 1$, the general solution is

$$T(t) = c_1 e^{-t} + c_2 t e^{-t}.$$

When $\Delta = 4 - 4\lambda < 0$, i.e. $\lambda > 1$, the general solution is

$$T(t) = c_1 e^{-t} \cos\sqrt{\lambda - 1}t + c_2 e^{-t} \sin\sqrt{\lambda - 1}t.$$

Question 3

Solve the equation

$$\begin{cases} u_{tt} = c^2 u_{xx}, \quad 0 < x < \pi, \quad t > 0, \\ u_x(0,t) = u_x(\pi,t) = 0, \quad t > 0, \\ u(x,0) = 0, \quad u_t(x,0) = \cos^2 x, \quad 0 < x < \pi. \end{cases}$$

Question 3

Solve the equation $u(x,0) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos nx$

$$\begin{cases} u_{tt} = c^2 u_{xx}, \quad 0 < x < \pi, \quad t > 0, \\ u_x(0,t) = u_x(\pi,t) = 0, \quad t > 0, \\ u(x,0) = 0, \quad u_t(x,0) = \cos^2 x, \quad 0 < x < \pi. \end{cases}$$

Using the solution formula for the Neumann problem on the finite interval $(0,\pi)$, we can write the solution as

$$u(x,t) = \frac{A_0}{2} + \frac{B_0}{2} t + \sum_{n=1}^{\infty} \left(A_n \cos nct + B_n \sin nct\right) \cos nx.$$

The initial condition $u(x,0) = 0$ implies $A_n = 0$ for all n, and the condition:

$$u_t(x,0) = \cos^2 x = \frac{1+\cos 2x}{2} = \frac{B_0}{2} + \sum_{n=1}^{\infty} B_n nc \cos nx,$$

we find $B_0 = 1$, $B_2 = \frac{1}{4c}$, and $B_n = 0$ for all the others n. Therefore, we have the solution:

$$u(x,t) = \frac{t}{2} + \frac{1}{4c} \sin 2ct \cos 2x. \qquad \blacksquare$$

Question 4

Solve the equation

$$\begin{cases} u_{tt} = c^2 u_{xx}, \quad 0 < x < \pi, \quad t > 0, \\ u_x(0,t) = u_x(\pi,t) = 0, \quad t > 0, \\ u(x,0) = x - 1, \quad u_t(x,0) = \sin^2 x, \quad 0 < x < \pi. \end{cases}$$

Question 4

Solve the equation

$$\begin{cases} u_{tt} = c^2 u_{xx}, \quad 0 < x < \pi, \quad t > 0, \\ u_x(0,t) = u_x(\pi,t) = 0, \quad t > 0, \\ u(x,0) = x - 1, \quad u_t(x,0) = \sin^2 x, \quad 0 < x < \pi. \end{cases}$$

Using the solution formula for the Neumann problem on $(0,\pi)$, we can write the solution as

$$u(x,t) = \frac{A_0}{2} + \frac{B_0}{2}t + \sum_{n=1}^{\infty} \left(A_n \cos nct + B_n \sin nct\right) \cos nx.$$

The initial condition $u(x,0) = x - 1$ implies that

$$x - 1 = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos nx.$$

So that A_n are the coefficients of the Fourier cosine series of $x - 1$, which can be calculated as

$$A_n = \frac{2}{\pi}\int_0^{\pi} (x-1)\cos nx dx = \begin{cases} \frac{2}{\pi}\int_0^{\pi} (x-1) dx, & n = 0 \\ \frac{2}{\pi}\int_0^{\pi} (x-1)\cos nx dx, & n \geq 1 \end{cases}$$

Question 4

$$A_n = \frac{2}{\pi}\int_0^{\pi}(x-1)\cos nx dx = \begin{cases} \frac{2}{\pi}\int_0^{\pi}(x-1)dx, & n=0 \\ \\ \frac{2}{\pi}\int_0^{\pi}(x-1)\cos nx dx, & n \geq 1 \end{cases}$$

$$= \begin{cases} \frac{2}{\pi}\left(\frac{x^2}{2}-x\right)\Big|_0^{\pi}, & n=0 \\ \\ \frac{2}{\pi}\left[\left(\frac{1}{n}(x-1)\sin nx\right)\Big|_0^{\pi} - \int_0^{\pi}\frac{1}{n}\sin nx dx\right], & n \geq 1. \end{cases}$$

After simplification, we get

$$A_n = \begin{cases} \pi - 2, & n=0 \\ \\ \frac{2}{n^2\pi}\left[(-1)^n - 1\right], & n \geq 1. \end{cases}$$

Question 4

The initial condition $u_t(x,0) = \sin^2 x$ implies that

$$u_t(x,0) = \sin^2 x = \frac{1-\cos 2x}{2} = \frac{B_0}{2} + \sum_{n=1}^{\infty} B_n nc \cos nx,$$

we find $B_0 = 1$, $B_2 = -\frac{1}{4c}$, and $B_n = 0$ for all the others n. Therefore, we have the solution:

$$u(x,t) = \frac{\pi-2}{2} + \frac{t}{2} - \frac{4}{\pi}\cos ct \cos x - \frac{1}{4c}\sin 2ct \cos 2x + \sum_{n=3}^{\infty} \frac{2}{n^2\pi}\left[(-1)^n - 1\right]\cos nct \cos nx,$$

which can be rewritten as

$$u(x,t) = \frac{\pi-2}{2} + \frac{t}{2} - \frac{4}{\pi}\cos ct \cos x - \frac{1}{4c}\sin 2ct \cos 2x - \sum_{k=1}^{\infty} \frac{4}{(2k+1)^2\pi}\cos(2k+1)ct \cos(2k+1)x.$$

Question 5

Let $\phi(x) = \cos x$ for $0 \le x \le \pi/2$.

(a) Calculate its Fourier sine series.

(b) Show that

$$\sum_{n=0}^{\infty} \frac{(-1)^n (n+\frac{1}{2})}{(n+\frac{1}{4})(n+\frac{3}{4})} = \frac{\sqrt{2}}{2}\pi.$$

Question 5

(a) Let

$$\cos x = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{\frac{\pi}{2}}, \quad 0 < x < \pi/2.$$

The coefficient is given by

$$A_n = \frac{2}{\frac{\pi}{2}} \int_0^{\pi/2} \cos x \sin \frac{n\pi x}{\frac{\pi}{2}} dx, \quad n = 1, 2, \cdots. \tag{3}$$

We know

$$\begin{aligned} A_n = \frac{4}{\pi} \int_0^{\pi/2} \cos x \sin \frac{n\pi x}{\frac{\pi}{2}} dx &= \frac{4}{\pi} \int_0^{\pi/2} \left[\frac{\sin(2n+1)x + \sin(2n-1)x}{2} \right] dx \\ &= \frac{4}{\pi} \left[-\frac{\cos(2n+1)x}{2(2n+1)} \Big|_{x=0}^{x=\frac{\pi}{2}} - \frac{\cos(2n-1)x}{2(2n-1)} \Big|_{x=0}^{x=\frac{\pi}{2}} \right] \\ &= \frac{4}{\pi} \left[\frac{1}{2(2n+1)} + \frac{1}{2(2n-1)} \right] \\ &= \frac{4}{\pi} \frac{2n}{(2n+1)(2n-1)} = \frac{8n}{(2n+1)(2n-1)\pi}. \end{aligned}$$

Thus,

$$\cos x = \sum_{n=1}^{\infty} \frac{8n}{(2n+1)(2n-1)\pi} \sin(2nx), \quad 0 < x < \frac{\pi}{2}.$$

pointwise convergence

Question 5

(b) Taking $x = \frac{\pi}{4}$ and using the Fourier convergence theorem, we find that

$$\frac{\sqrt{2}}{2} = \sum_{n=1}^{\infty} \frac{8n}{(2n+1)(2n-1)\pi} \sin\left(\frac{n}{2}\pi\right).$$

When n is even,

$$\sin\left(\frac{n}{2}\pi\right) = 0.$$

When n is odd or $n = 2k+1$ $(k \geq 0)$, we have

$$\begin{aligned}\frac{8n}{(2n+1)(2n-1)\pi} \sin\left(\frac{n}{2}\pi\right) &= \frac{16k+8}{(4k+3)(4k+1)\pi} \sin\left(\frac{2k+1}{2}\pi\right) \\ &= \frac{16k+8}{(4k+3)(4k+1)\pi}(-1)^k.\end{aligned}$$

This is equivalent to

$$\sum_{n=0}^{\infty} \frac{(-1)^n(n+\frac{1}{2})}{(n+\frac{1}{4})(n+\frac{3}{4})} = \frac{\sqrt{2}}{2}\pi.$$

Question 6

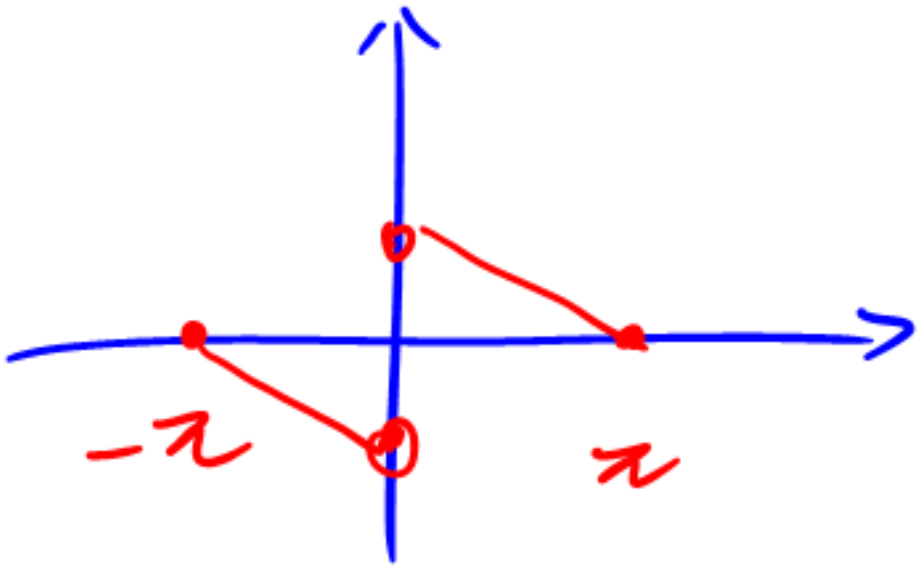
A sawtooth wave is denoted by

$$f(x) = \begin{cases} -(\pi + x)/2, & -\pi \le x < 0, \\ (\pi - x)/2, & 0 < x \le \pi. \end{cases}$$

Show that

$$f(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n},$$

and discuss its pointwise convergence.



Question 6

Notice that the function is odd in $(-\pi, \pi)$. Therefore,

$$A_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx dx = 0, \quad n \geq 0,$$

and

$$\begin{aligned} B_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx dx = \frac{2}{\pi}\int_{0}^{\pi} \frac{\pi - x}{2}\sin nx dx \\ &= \frac{1}{n\pi}(x-\pi)|_0^{\pi} - \frac{1}{n^2\pi}\sin nx|_0^{\pi} = \frac{1}{n}. \end{aligned}$$

Therefore, we have

$$f(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n}.$$

1 2

By the pointwise convergence, we know that the series converges to $f(x)$ for $x \neq 0$, and at $x = 0$, converges to 0.

Question 7

Let $f(x) = |\cos x|$ on the interval $(0, \pi)$.

(a) Find the Fourier cosine series of $f(x)$.

(b) Find the sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}.$$

Question 7

(a) Let

$$|\cos x| = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos nx, \quad 0 < x < \pi.$$

The coefficient is given by

$$\begin{aligned} A_n &= \frac{2}{\pi}\int_0^{\pi} |\cos x| \cos(nx) dx \\ &= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \cos x \cos(nx) dx - \frac{2}{\pi}\int_{\frac{\pi}{2}}^{\pi} \cos x \cos(nx) dx, \quad n = 0, 1, 2, \cdots . \end{aligned}$$

If $n = 0$, it is easy to get $A_0 = \frac{4}{\pi}$.

If $n \neq 0$, we know

$$\cos x \cos(nx) = \frac{1}{2}\left[\cos(n+1)x + \cos(n-1)x\right].$$

When $n = 1$, $A_1 = 0$;

When $n > 1$,

$$A_n = \frac{1}{\pi}\int_0^{\frac{\pi}{2}} \left[\cos(n+1)x + \cos(n-1)x\right] dx - \frac{1}{\pi}\int_{\frac{\pi}{2}}^{\pi} \left[\cos(n+1)x + \cos(n-1)x\right] dx$$

Question 7

(a) When $n > 1$,

$$
\begin{aligned}
A_n =& \frac{1}{\pi}\int_0^{\frac{\pi}{2}} \left[\cos(n+1)x + \cos(n-1)x\right] dx - \frac{1}{\pi}\int_{\frac{\pi}{2}}^{\pi} \left[\cos(n+1)x + \cos(n-1)x\right] dx \\
=& \frac{1}{\pi}\left[\frac{1}{n+1}\sin(n+1)\frac{\pi}{2} + \frac{1}{n-1}\sin(n-1)\frac{\pi}{2}\right] \\
&+ \frac{1}{\pi}\left[\frac{1}{n+1}\sin(n+1)\frac{\pi}{2} + \frac{1}{n-1}\sin(n-1)\frac{\pi}{2}\right] \\
=& \frac{2}{\pi}\left[\frac{1}{n+1}\sin(n+1)\frac{\pi}{2} + \frac{1}{n-1}\sin(n-1)\frac{\pi}{2}\right] = \frac{2}{\pi}\left[\frac{1}{n+1}\cos\frac{n\pi}{2} - \frac{1}{n-1}\cos\frac{n\pi}{2}\right] \\
=& \frac{4}{(1-n^2)\pi}\cos\frac{n\pi}{2}.
\end{aligned}
$$

Or

$$
A_{2k-1} = 0, \qquad A_{2k} = \frac{4(-1)^{k+1}}{(4k^2-1)\pi},
$$

where $k = 1, 2 \cdots$. Thus

$$
|\cos x| = \frac{2}{\pi} + \sum_{k=1}^{\infty} \frac{4(-1)^{k+1}}{(4k^2-1)\pi}\cos(2kx), \quad 0 < x < \pi.
$$

(b) Taking $x = \pi/2$ and using the Fourier convergence theorem, we find that

Question 7

(b) Taking $x = \pi/2$ and using the Fourier convergence theorem, we find that

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{1}{2}.$$